

1. List all customers from USA.

Query:

```
SELECT *
FROM customers
WHERE country = 'USA';
```

Output:

	actitle acter varying (30)	address character varying (60)	city character varying (15)	region character varying (15)	postalcode character varying (10)	country character varying (15)	phone character varying (24)	fax char
1	keting Manager	2732 Baker Blvd.	Eugene	OR	97403	USA	(503) 555-7555	[nul
2	is Representative	City Center Plaza 516 Main ...	Elgin	OR	97827	USA	(503) 555-6874	(50
3	keting Manager	12 Orchestra Terrace	Walla Walla	WA	99362	USA	(509) 555-7969	(50
4	er	87 Polk St. Suite 5	San Francisco	CA	94117	USA	(415) 555-5938	[nul
5	is Manager	89 Chiaroscuro Rd.	Portland	OR	97219	USA	(503) 555-9573	(50
6	is Representative	2743 Bering St.	Anchorage	AK	99508	USA	(907) 555-7584	(90
7	istant Sales Representat...	2817 Milton Dr.	Albuquerque	NM	87110	USA	(505) 555-5939	(50
8	is Representative	187 Suffolk Ln.	Boise	ID	83720	USA	(208) 555-8097	[nul

2. List all products where UnitPrice is greater than 20.

Query

```
SELECT *
FROM products
WHERE unitprice > 20;
```

Output

	productid [PK] integer	productname character varying (40)	supplierid integer	categoryid integer	quantityperunit character varying (20)	unitprice numeric (19,4)	unitsinstock smallint	unitsonorder smallint	reorderlevel smallint	discontinued smallint
1	4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22.0000	53	0	0	0
2	5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.3500	0	0	0	1
3	6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25.0000	120	0	25	0
4	7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30.0000	15	0	10	0
5	8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40.0000	6	0	0	0
6	9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97.0000	29	0	0	1
7	10	Ikura	4	8	12 - 200 ml jars	31.0000	31	0	0	0
8	11	Queso Cabrales	5	4	1 kg pkg.	21.0000	22	30	30	0

3. List all orders placed after 1997-01-01.

Query

```
SELECT *
FROM orders
WHERE orderdate > '1997-01-01';
```

Output

	orderid [PK] integer	customerid character (5)	employeeid integer	orderdate timestamp without time zone	requireddate timestamp without time zone	shippeddate timestamp without time zone	shipvia integer	freight numeric (19,4)	shipname character vary
1	10402	ERNSH	8	1997-01-02 00:00:00	1997-02-13 00:00:00	1997-01-10 00:00:00	2	67.8800	Ernst Handel
2	10403	ERNSH	4	1997-01-03 00:00:00	1997-01-31 00:00:00	1997-01-09 00:00:00	3	73.7900	Ernst Handel
3	10404	MAGAA	2	1997-01-03 00:00:00	1997-01-31 00:00:00	1997-01-08 00:00:00	1	155.9700	Magazzini Ali
4	10405	LINOD	1	1997-01-06 00:00:00	1997-02-03 00:00:00	1997-01-22 00:00:00	1	34.8200	LINO-Delicate
5	10406	QUEEN	7	1997-01-07 00:00:00	1997-02-18 00:00:00	1997-01-13 00:00:00	1	108.0400	Queen Cozinha
6	10407	OTTIK	2	1997-01-07 00:00:00	1997-02-04 00:00:00	1997-01-30 00:00:00	2	91.4800	Ottillies Käsel
7	10408	FOLIG	8	1997-01-08 00:00:00	1997-02-05 00:00:00	1997-01-14 00:00:00	1	11.2600	Folies gourm.
8	10409	OCEAN	3	1997-01-09 00:00:00	1997-02-06 00:00:00	1997-01-14 00:00:00	1	29.8300	Océano Atlánt

4. Display customers ordered by Country and then CompanyName.

Query

```
SELECT *  
FROM customers  
ORDER BY country, companyname; -- for customers with same country we sort using companyname
```

Output

	customerid [PK] character (5)	companyname character varying (40)	contactname character varying (30)	contacttitle character varying (30)	address character varying (60)	city character varying (15)	region character varying (15)
1	GREAL	Great Lakes Food Market	Howard Snyder	Marketing Manager	2732 Baker Blvd.	Eugene	OR
2	HUNGC	Hungry Coyote Import Store	Yoshi Latimer	Sales Representative	City Center Plaza 516 Main ...	Elgin	OR
3	LAZYK	Lazy K Kountry Store	John Steel	Marketing Manager	12 Orchestra Terrace	Walla Walla	WA
4	LETSS	Let's Stop N Shop	Jaime Yorres	Owner	87 Polk St. Suite 5	San Francisco	CA
5	LONEP	Lonesome Pine Restaurant	Fran Wilson	Sales Manager	89 Chiaroscuro Rd.	Portland	OR
6	OLDWO	Old World Delicatessen	Rene Phillips	Sales Representative	2743 Bering St.	Anchorage	AK
7	RATTC	Rattlesnake Canyon Grocery	Paula Wilson	Assistant Sales Representat...	2817 Milton Dr.	Albuquerque	NM
8	SAVEA	Save-a-lot Markets	Jose Pavarotti	Sales Representative	187 Suffolk Ln.	Boise	ID

5. List products ordered by highest UnitPrice first.

Query

```
SELECT *  
FROM products  
ORDER BY unitprice DESC;
```

Output

	productid [PK] integer	productname character varying (40)	supplierid integer	categoryid integer	quantityperunit character varying (20)	unitprice numeric (19,4)	unitsinstock smallint	unitsonorder smallint	reorderlevel smallint	discontinued smallint
1	38	Côte de Blaye	18	1	12 - 75 cl bottles	263.5000	17	0	15	0
2	29	Thüringer Rostbratwurst	12	6	50 bags x 30 sausgs.	123.7900	0	0	0	1
3	9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97.0000	29	0	0	1
4	20	Sir Rodney's Marmalade	8	3	30 gift boxes	81.0000	40	0	0	0
5	18	Carnarvon Tigers	7	8	16 kg pkg.	62.5000	42	0	0	0
6	59	Raclette Courdavault	28	4	5 kg pkg.	55.0000	79	0	0	0
7	51	Manjimup Dried Apples	24	7	50 - 300 g pkgs.	53.0000	20	0	10	0
8	62	Tarte au sucre	29	3	48 pies	49.3000	17	0	0	0

6. Count how many customers are there in each country.

Query

```
SELECT country,  
COUNT(*) AS customer_per_country  
FROM customers  
GROUP BY country
```

Output

	country character varying (15)	customer_per_country bigint
1	Argentina	3
2	Spain	5
3	Switzerland	2
4	Italy	3
5	Venezuela	4
6	Belgium	2
7	Norway	1
8	Sweden	2
9	USA	13

7. Find the number of products in each category.

Query

```
SELECT categoryid,  
COUNT(*) AS products_per_category  
FROM products  
GROUP BY categoryid;
```

Output

	categoryid integer	products_per_category bigint
1	8	12
2	7	5
3	1	12
4	5	7
5	4	10
6	2	12
7	6	6
8	3	13

8. Find the total number of orders handled by each employee.

Query

```
SELECT employeeid, COUNT(*) AS orders_per_employee
FROM orders
GROUP BY employeeid;
```

Output

	employeeid integer	orders_per_employee bigint
1	8	104
2	7	72
3	9	43
4	1	123
5	5	42

9. Find the average freight amount for each customer.

Query

```
SELECT customerid,
AVG(freight) AS avg_freight_per_customer
FROM orders
GROUP BY customerid;
```

Output

	customerid character (5) 🔒	avg_freight_per_customer numeric 🔒
1	TOMSP	20.9950000000000000
2	LONEP	9.9825000000000000
3	OLDWO	98.3530000000000000
4	WARTH	54.8320000000000000
5	MAGAA	46.9750000000000000

10. Find the maximum unit price in each category.

Query

```
SELECT categoryid,  
MAX(unitprice) AS max_unit_price_per_category  
FROM products  
GROUP BY categoryid;
```

Output

	categoryid integer 🔒	max_unit_price_per_category numeric 🔒
1	8	62.5000
2	7	53.0000
3	1	263.5000
4	5	38.0000
5	4	55.0000
6	2	43.9000
7	6	123.7900
8	3	81.0000

11. Show countries having more than 5 customers.

Query

```
SELECT country AS country_with_more_than_5_customers  
FROM customers  
GROUP BY country  
HAVING COUNT(customerid) > 5;
```

Output

	country_with_more_than_5_customers character varying (15) 🔒
1	USA
2	France
3	Brazil
4	UK
5	Germany

12. Show employees who handled more than 50 orders.

Query

```
SELECT employeeid AS employeeid_with_more_than_50_orders
FROM orders
GROUP BY employeeid
HAVING COUNT(orderid) > 50;
```

Output

	employeeid_with_more_than_50_orders integer 🔒
1	8
2	7
3	1
4	2
5	4
6	6
7	3

13. Show customers whose average freight is greater than 50.

Query

```
SELECT customerid
FROM orders
GROUP BY customerid
HAVING AVG(freight) > 50;
```

Output

	customerid character (5) 🔒
1	OLDWO
2	WARTH
3	QUEEN
4	MORGK
5	RATTC
6	FRANK
7	PRINI

14. Show categories where the average product price is greater than 30.

Query

```
SELECT categoryid  
FROM products  
GROUP BY categoryid  
HAVING avg(unitprice) > 30;
```

Output

	categoryid integer 🔒
1	7
2	1
3	6

15. Show ship countries having more than 20 orders.

Query

```
SELECT shipcountry, count(shipcountry)  
FROM orders  
GROUP BY shipcountry  
HAVING COUNT(shipcountry) > 20;
```

Output

	shipcountry character varying (15)	count bigint
1	Spain	23
2	Italy	28
3	Venezuela	46
4	Sweden	37
5	USA	122

16. List each order with customer company name.

Query

```
SELECT o.*, c.companyname
FROM orders o
JOIN customers c ON o.customerid = c.customerid;
```

Output

	shipaddress character varying (60)	shipcity character varying (15)	shipregion character varying (15)	shippostalcode character varying (10)	shipcountry character varying (15)	companyname character varying (40)
1	59 rue de l'Abbaye	Reims	[null]	51100	France	Vins et alcools Chevalier
2	Luisenstr. 48	Münster	[null]	44087	Germany	Toms Spezialitäten
3	Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	Hanari Carnes
4	2, rue du Commerce	Lyon	[null]	69004	France	Victuailles en stock
5	Boulevard Tirou, 255	Charleroi	[null]	B-6000	Belgium	Suprêmes délices
6	Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	Hanari Carnes

17. List each order with employee first name and last name.

Query

```
SELECT o.*, e.firstname,e.lastname
FROM orders o
JOIN employees e ON o.employeeid = e.employeeid;
```

Output

	shipaddress character varying (60)	shipcity character varying (15)	shipregion character varying (15)	shippostalcode character varying (10)	shipcountry character varying (15)	firstname character varying (10)	lastname character varying (20)
1	59 rue de l'Abbaye	Reims	[null]	51100	France	Steven	Buchanan
2	Luisenstr. 48	Münster	[null]	44087	Germany	Michael	Suyama
3	Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	Margaret	Peacock
4	2, rue du Commerce	Lyon	[null]	69004	France	Janet	Leverling
5	Boulevard Tirou, 255	Charleroi	[null]	B-6000	Belgium	Margaret	Peacock
6	Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	Janet	Leverling

18. List products with their category name.

Query

```
SELECT p.*, c.categoryname
FROM products p
JOIN categories c ON p.categoryid = c.categoryid;
```

Output

	productname character varying (40)	supplierid integer	categoryid integer	quantityunit character varying (20)	unitprice numeric (19,4)	unitsinstock smallint	unitsonorder smallint	reorderlevel smallint	discontinued smallint	categoryname character varying (15)
1		1	1	10 boxes x 20 bags	18.0000	39	0	10	0	Beverages
2		1	1	24 - 12 oz bottles	19.0000	17	40	25	0	Beverages
3	Syrup	1	2	12 - 550 ml bottles	10.0000	13	70	25	0	Condiments
4	on's Cajun Seasoning	2	2	48 - 6 oz jars	22.0000	53	0	0	0	Condiments
5	on's Gumbo Mix	2	2	36 boxes	21.3500	0	0	0	1	Condiments
6	's Boysenberry Spread	3	2	12 - 8 oz jars	25.0000	120	0	25	0	Condiments

19. List products with supplier company name.

Query

```
SELECT p.*, s.companyname
FROM products p
JOIN suppliers s ON p.supplierid = s.supplierid;
```

Output

	productname character varying (40)	supplierid integer	categoryid integer	quantityunit character varying (20)	unitprice numeric (19,4)	unitsinstock smallint	unitsonorder smallint	reorderlevel smallint	discontinued smallint	companyname character varying (40)
1		1	1	10 boxes x 20 bags	18.0000	39	0	10	0	Exotic Liquids
2		1	1	24 - 12 oz bottles	19.0000	17	40	25	0	Exotic Liquids
3		1	2	12 - 550 ml bottles	10.0000	13	70	25	0	Exotic Liquids
4	ning	2	2	48 - 6 oz jars	22.0000	53	0	0	0	New Orleans Cajun Delights
5		2	2	36 boxes	21.3500	0	0	0	1	New Orleans Cajun Delights
6	pread	3	2	12 - 8 oz jars	25.0000	120	0	25	0	Grandma Kelly's Homestead

20. List orders with shipper company name.

Query

```
SELECT o.*, sh.companyname
FROM orders o
JOIN shippers sh ON o.shipvia = sh.shipperid;
```

Output

		shipaddress character varying (60)	shipcity character varying (15)	shipregion character varying (15)	shippostalcode character varying (10)	shipcountry character varying (15)	companyname character varying (40)
1	Chavalier	59 rue de l'Abbaye	Reims	[null]	51100	France	Federal Shipping
2		Luisenstr. 48	Münster	[null]	44087	Germany	Speedy Express
3		Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	United Package
4	Chavalier	2, rue du Commerce	Lyon	[null]	69004	France	Speedy Express
5		Boulevard Tirou, 255	Charleroi	[null]	B-6000	Belgium	United Package
6		Rua do Paço, 67	Rio de Janeiro	RJ	05454-876	Brazil	United Package

21. Find total orders per customer and display customer company name.

Query

```
SELECT c.companyname,  
COUNT(o.orderid) as total_orders  
FROM customers c  
JOIN orders o ON c.customerid = o.customerid  
GROUP BY c.companyname;
```

Output

	companyname character varying (40)	total_orders bigint
1	Wilman Kala	7
2	Magazzini Alimentari Riuniti	10
3	The Cracker Box	3
4	Wellington Importadora	9
5	Let's Stop N Shop	4
6	Furia Bacalhau e Frutos do Mar	8
7	Toms Spezialitäten	6

22. Find total products supplied by each supplier.

Query

```
SELECT s.supplierid, s.companyname,  
COUNT(*) as total_products  
FROM products p  
JOIN suppliers s ON p.supplierid = s.supplierid  
GROUP BY s.supplierid;
```

Output

	supplierid [PK] integer	companyname character varying (40)	total_products bigint
1	29	Forêts d'érables	2
2	4	Tokyo Traders	3
3	10	Refrescos Americanas LTDA	1
4	6	Mayumi's	3
5	14	Formaggi Fortini s.r.l.	3

23. Find average product price per category with category name.

Query

```
SELECT c.categoryid, c.categoryname,  
AVG(p.unitprice) AS avg_price  
FROM categories c  
JOIN products p ON c.categoryid = p.categoryid  
GROUP BY c.categoryid;
```

Output

	categoryid [PK] integer	categoryname character varying (15)	avg_price numeric
1	4	Dairy Products	28.7300000000000000
2	6	Meat/Poultry	54.0066666666666667
3	2	Condiments	23.0625000000000000
4	7	Produce	32.3700000000000000
5	3	Confections	25.1600000000000000
6	1	Beverages	37.9791666666666667
7	5	Grains/Cereals	20.2500000000000000
8	8	Seafood	20.6825000000000000

24. Find total freight per customer and order by highest total freight.

```
SELECT c.customerid, c.companyname,  
SUM(o.freight) AS total_freight  
FROM customers c  
JOIN orders o ON c.customerid = o.customerid  
GROUP BY c.customerid  
ORDER BY total_freight DESC;
```

	customerid [PK] character (5)	companyname character varying (40)	total_freight numeric
1	SAVEA	Save-a-lot Markets	6683.7000
2	ERNSH	Ernst Handel	6205.3900
3	QUICK	QUICK-Stop	5605.6300
4	HUNGO	Hungry Owl All-Night Grocers	2755.2400

25. Find employees who handled more than 25 orders.

Query

```
SELECT e.employeeid, e.firstname, e.lastname,
COUNT(*) AS total_orders
FROM employees e
JOIN orders o ON e.employeeid = o.employeeid
GROUP BY e.employeeid
HAVING COUNT(*) > 25;
```

Output

	employeeid [PK] integer	firstname character varying (10)	lastname character varying (20)	total_orders bigint
1	4	Margaret	Peacock	156
2	6	Michael	Suyama	67
3	2	Andrew	Fuller	96
4	7	Robert	King	72
5	9	Anne	Dodsworth	43

26. Find total sales amount per order.

Query

```
SELECT o.orderid,
SUM(od.unitprice * od.quantity * (1 - od.discount)) AS total_sales
FROM orders o
JOIN order_details od ON o.orderid = od.orderid
GROUP BY o.orderid;
```

Output

	orderid [PK] integer	total_sales double precision
1	11038	732.5999997258186
2	10782	12.5
3	10725	287.8
4	10423	1020
5	10518	4150.05

27. Find total sales amount per customer.

Query

```
SELECT c.customerid, c.companyname,  
SUM(od.unitprice * od.quantity * (1 - od.discount)) AS total_sales  
FROM customers c  
JOIN orders o ON c.customerid = o.customerid  
JOIN order_details od ON o.orderid = od.orderid  
GROUP BY c.customerid;
```

Output

	customerid [PK] character (5) 	companyname character varying (40) 	total_sales double precision 
1	TOMSP	Toms Spezialitäten	4778.139997901022
2	LONEP	Lonesome Pine Restaurant	4258.6
3	OLDWO	Old World Delicatessen	15177.462481805682
4	WARTH	Wartian Herkku	15648.702469434587
5	MAGAA	Magazzini Alimentari Riuniti	7176.214986810461

28. Find top 10 products by total quantity sold.

Query

```
SELECT p.productid, p.productname, SUM(od.quantity) AS total_quantity  
FROM products p  
JOIN order_details od ON p.productid = od.productid  
GROUP BY p.productid  
ORDER BY total_quantity DESC  
LIMIT 10;
```

Output

	productid [PK] integer	productname character varying (40)	total_quantity bigint
1	60	Camembert Pierrot	1577
2	59	Raclette Courdavault	1496
3	31	Gorgonzola Telino	1397
4	56	Gnocchi di nonna Alice	1263
5	16	Pavlova	1158
6	75	Rhönbräu Klosterbier	1155
7	24	Guaraná Fantástica	1125
8	40	Boston Crab Meat	1103
9	62	Tarte au sucre	1083
10	2	Chang	1057

29. Find categories whose total sales are greater than 50000.

Query

```
SELECT c.categoryid, c.categoryname,
SUM(od.unitprice * od.quantity) AS total_sales
FROM order_details od
JOIN products p ON p.productid = od.productid
JOIN categories c ON c.categoryid = p.categoryid -- here we are joining a 3rd table
GROUP BY c.categoryid
HAVING SUM(od.unitprice * od.quantity) > 50000;
```

Output

	categoryid [PK] integer	categoryname character varying (15)	total_sales numeric
1	4	Dairy Products	251330.5000
2	6	Meat/Poultry	178188.8000
3	2	Condiments	113694.7500
4	3	Confections	177099.1000
5	5	Grains/Cereals	100726.8000
6	7	Produce	105268.6000
7	1	Beverages	286526.9500
8	8	Seafood	141623.0900

30. Find employees whose total sales are greater than 100000.

Query

```
SELECT e.employeeid, e.firstname, e.lastname,
SUM(od.unitprice * od.quantity) AS total_sales
FROM employees e
JOIN orders o ON e.employeeid = o.employeeid
JOIN order_details od ON od.orderid = o.orderid
```

GROUP BY e.employeeid
HAVING SUM(od.unitprice * od.quantity) > 100000;

Output

	employeeid [PK] integer	firstname character varying (10)	lastname character varying (20)	total_sales numeric
1	4	Margaret	Peacock	250187.4500
2	2	Andrew	Fuller	177749.2600
3	7	Robert	King	141295.9900
4	3	Janet	Leverling	213051.3000
5	1	Nancy	Davolio	202143.7100
6	8	Laura	Callahan	133301.0300

31. Find total sales per country based on customer country.

Query

SELECT c.country,
SUM(od.unitprice * od.quantity) AS total_sales
FROM customers c
JOIN orders o ON c.customerid = o.customerid
JOIN order_details od ON od.orderid = o.orderid
GROUP BY c.country;

Output

	country character varying (15)	total_sales numeric
1	Argentina	8119.1000
2	Spain	19431.8900
3	Switzerland	32919.5000
4	Italy	16705.1500
5	Venezuela	60814.8900

32. Find suppliers whose products generated sales above 30000.

Query

SELECT s.supplierid, s.companyname,
SUM(od.unitprice * od.quantity) AS total_sales
FROM suppliers s

```

JOIN products p ON s.supplierid = p.supplierid
JOIN order_details od ON od.productid = p.productid
GROUP BY s.supplierid
HAVING SUM(od.unitprice * od.quantity) > 30000;

```

Output

	supplierid [PK] integer	companyname character varying (40)	total_sales numeric
1	29	Forêts d'érables	66266.7000
2	4	Tokyo Traders	33533.2000
3	14	Formaggi Fortini s.r.l.	51082.5000
4	2	New Orleans Cajun Delights	33351.9500
5	11	Heli Süßwaren GmbH & Co. KG	40818.0000
6	7	Pavlova, Ltd.	115386.0500

33. Find customers who placed more than 10 orders and sort by order count descending.

Query

```

SELECT c.customerid, c.companyname, --
COUNT(*) AS total_orders
FROM customers c
JOIN orders o ON c.customerid = o.customerid
GROUP BY c.customerid
HAVING COUNT(*) > 10
ORDER BY total_orders DESC;

```

Output

	customerid [PK] character (5)	companyname character varying (40)	total_orders bigint
1	SAVEA	Save-a-lot Markets	31
2	ERNSH	Ernst Handel	30
3	QUICK	QUICK-Stop	28
4	FOLKO	Folk och få HB	19
5	HUNGO	Hungry Owl All-Night Groc...	19

34. Find monthly order count for each year and month.

Query

```

SELECT TO_CHAR(orderdate, 'YYYY-MM') AS year_month,
-- using TO_CHAR() to strip year and month
COUNT(*) AS order_count
FROM orders

```



```
GROUP BY TO_CHAR(orderdate, 'YYYY-MM')
ORDER BY year_month;
```

Output

	year_month text	order_count bigint
1	1996-07	22
2	1996-08	25
3	1996-09	23
4	1996-10	26
5	1996-11	25

35. Find monthly sales amount ordered by year and month.

Query

```
SELECT TO_CHAR(o.orderdate, 'YYYY-MM') AS year_month,
-- using TO_CHAR() to strip year and month
SUM(od.unitprice * od.quantity * (1 - od.discount)) AS total_sales
FROM orders o
JOIN order_details od ON od.orderid = o.orderid
GROUP BY TO_CHAR(o.orderdate, 'YYYY-MM')
ORDER BY year_month;
```

Output

	year_month text	total_sales double precision
1	1996-07	27861.89495651722
2	1996-08	25485.27498658151
3	1996-09	26381.399988487363
4	1996-10	37515.72491448447
5	1996-11	45600.04494312592